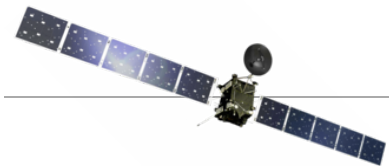


Rosetta (spacecraft)



Artist's illustration of *Rosetta*

Mission type	Comet orbiter/lander
Operator	ESA
COSPAR ID	2004-006A (https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2004-006A)
SATCAT no.	28169
Website	esa.int/rosetta (http://esa.int/rosetta)
Mission duration	12 years, 6 months and 28 days
Spacecraft properties	
Manufacturer	Astrium
Launch mass	<i>Combined:</i> 3,000 kg (6,600 lb) <i>Orbiter:</i> 2,900 kg (6,400 lb) <i>Lander:</i> 100 kg (220 lb) ^[1]
Dry mass	<i>Orbiter:</i> 1,230 kg (2,710 lb)
Payload mass	<i>Orbiter:</i> 165 kg (364 lb) <i>Lander:</i> 27 kg (60 lb)
Dimensions	2.8 × 2.1 × 2 m (9.2 × 6.9 × 6.6 ft)
Power	850 watts at 3.4 AU ^[2]
Start of mission	
Launch date	2 March 2004, 07:17:51 UTC ^[3]
Rocket	Ariane 5G+ V-158
Launch site	Kourou ELA-3
Contractor	Arianespace
End of mission	
Disposal	Deorbited

Last contact	30 September 2016, 10:39:28 <u>UTC</u> <u>SCET</u>
Landing site	Sais, Ma'at region ^[4] 2 years, 55 days of operations at the comet

	Flyby of <u>Earth</u>
Closest approach	4 March 2005
Distance	1,954 km (1,214 mi)
	Flyby of <u>Mars</u>
Closest approach	25 February 2007
Distance	250 km (160 mi)
	Flyby of <u>Earth</u>
Closest approach	13 November 2007
Distance	5,700 km (3,500 mi)
	Flyby of <u>2867 Šteins</u>
Closest approach	5 September 2008
Distance	800 km (500 mi)
	Flyby of <u>Earth</u>
Closest approach	12 November 2009
Distance	2,481 km (1,542 mi)
	Flyby of <u>21 Lutetia</u>
Closest approach	10 July 2010
Distance	3,162 km (1,965 mi)
	<u>67P/Churyumov–Gerasimenko</u> orbiter
Spacecraft component	<i>Rosetta</i>
Orbital insertion	6 August 2014, 09:06 UTC ^[5]
	Orbital parameters
<u>Periapsis altitude</u>	29 km (18 mi) ^[6]
	<u>67P/Churyumov–Gerasimenko</u> lander
Spacecraft component	<i>Philae</i>
Landing date	12 November 2014, 17:32 UTC

Landing site Abydos	
Transponders	
Band	S band (low gain antenna) X band (high gain antenna)
Bandwidth	from 7.8-bit/s (S band)^[7] up to 91 kbit/s (X band)^[8]
Instruments	
	
Contemporary ESA insignia (2010s)	
Horizon 2000 (Science Programme)	

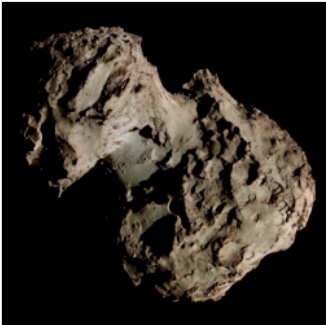
Rosetta was a space probe built by the European Space Agency that launched on 2 March 2004. Along with *Philae*, its lander module, *Rosetta* performed a detailed study of comet 67P/Churyumov–Gerasimenko (67P).^{[9][10]} During its journey to the comet, the spacecraft performed flybys of Earth, Mars, and the asteroids 21 Lutetia and 2867 Šteins.^{[11][12][13]} It was launched as the third cornerstone mission of the ESA's Horizon 2000 programme, after *SOHO* / *Cluster* and *XMM-Newton*.^[14] The total cost of the mission was about €1.3 billion (US\$1.8 billion).^[15]

On 6 August 2014, the spacecraft reached the comet and performed a series of manoeuvres to eventually orbit the comet at distances of 30 to 10 kilometres (19 to 6 mi).^[16] On 12 November, its lander module *Philae* performed the first successful landing on a comet,^[17] though its battery power ran out two days later.^[18] Communications with *Philae* were briefly restored in June and July 2015, but due to diminishing solar power, *Rosetta*'s communications module with the lander was turned off on 27 July 2016.^[19] On 30 September 2016, the *Rosetta* spacecraft ended its mission by hard-landing on the comet in its Ma'at region.^{[20][21]}

Name

The probe was named after the Rosetta Stone,^[22] a stele of Egyptian origin featuring a decree in three scripts. The lander was named after the Philae obelisk, which bears a bilingual Greek and Egyptian hieroglyphic inscription. A comparison of its hieroglyphs with those on the Rosetta Stone catalysed the deciphering of the Egyptian writing system. Similarly, it was hoped that these spacecraft would result in better understanding of comets and the early Solar System.^{[23][24]} In a more direct analogy to its namesake, the *Rosetta* spacecraft also carried a micro-etched pure nickel prototype of the Rosetta disc donated by the Long Now Foundation. The disc was inscribed with 6,500 pages of language translations.^{[25][26]}

Mission overview


 Comet Churyumov–Gerasimenko in March 2015 as imaged by *Rosetta*, enhanced colour

The spacecraft consisted of the *Rosetta* orbiter, which featured 12 instruments, and the *Philae* lander, with nine additional instruments.^[27] The *Rosetta* mission orbited Comet Churyumov–Gerasimenko for 17 months and was designed to complete the most detailed study of a comet ever attempted.

Rosetta was launched on 2 March 2004 from the Guiana Space Centre in Kourou, French Guiana, on an Ariane 5 rocket. In 2007, *Rosetta* made a Mars gravity assist (flyby).^[28] The spacecraft also performed two asteroid flybys:^[29] of 2867 Šteins in September 2008 and of 21 Lutetia in July 2010.^[30]

Rosetta reached Comet Churyumov–Gerasimenko in May 2014.^[31] It performed a series of manoeuvres to enter orbit between then and 6 August 2014,^[32] when it became the first

spacecraft to orbit a comet.^{[33][31][34]} *Rosetta*'s *Philae* lander successfully made the first soft landing on a comet nucleus when it touched down on Comet Churyumov–Gerasimenko on 12 November 2014.^{[35][36][37]}

Mission firsts

The *Rosetta* mission achieved many historic firsts.^[38] On its way to comet 67P, *Rosetta* passed through the main asteroid belt, and made the first European close encounter with several of these primitive objects. *Rosetta* was the first spacecraft to fly close to Jupiter's orbit using solar cells as its main power source.^[39] *Rosetta* was the first spacecraft to orbit a comet nucleus,^[40] and was the first spacecraft to fly alongside a comet as it headed towards the inner Solar System. Previous missions had conducted successful flybys of seven other comets.^[41] It became the first spacecraft to examine at close proximity the activity of a frozen comet as it is warmed by the Sun. Shortly after its arrival at 67P, the *Rosetta* orbiter dispatched the *Philae* lander for the first controlled touchdown on a comet nucleus. The robotic lander's instruments obtained the first images from a comet's surface and made the first on-site analysis of its composition.

History

Background

During the 1986 approach of Halley's Comet, international space probes were sent to explore the comet, most prominent among them being ESA's *Giotto*.^[42] After the probes returned valuable scientific information, it became obvious that follow-ons were needed that would shed more light on cometary composition and answer new questions.^[43]

Both ESA and NASA started cooperatively developing new probes. The NASA project was the Comet Rendezvous Asteroid Flyby (CRAF) mission.^[44] The ESA project was the follow-on Comet Nucleus Sample Return (CNSR) mission.^[45] Both missions were to share the *Mariner Mark II* spacecraft design, thus minimising costs. In 1992, after NASA cancelled CRAF due to budgetary limitations, ESA decided to develop a CRAF-style project on its own.^[46] By 1993 it was evident that the ambitious sample return mission was infeasible with the existing ESA budget, so the mission was redesigned and subsequently approved by the ESA, with the final flight plan resembling the cancelled CRAF mission: an asteroid flyby followed by a comet rendezvous with in-situ examination, including a lander.^[46] After the spacecraft launch, *Gerhard Schwehm* was named mission manager; he retired in March 2014.^[47]

The *Rosetta* mission included generational team management; this allowed mission continuity over the long period of the mission and for special knowledge to be maintained and passed on to future team members. In particular, several younger scientists were brought on as principal science investigators, and regular training sessions were conducted.^[16]

The spacecraft was controlled from the European Space Operations Centre (ESOC), in Darmstadt, Germany.^[48] The planning for the operation of the scientific payload, together with the data retrieval, calibration, archiving and distribution, was performed from the European Space Astronomy Centre (ESAC), in Villanueva de la Cañada, near Madrid, Spain.^[49] It has been estimated that in the decade preceding 2014, some 2,000 people assisted in the mission in some capacity.^[47]

Rosetta was built in a clean room according to COSPAR rules, but "sterilisation [was] generally not crucial since comets are usually regarded as objects where you can find prebiotic molecules, that is, molecules that are precursors of life, but not living microorganisms", according to Gerhard Schwehm, *Rosetta*'s project scientist.^[50]

Rosetta was set to be launched on 12 January 2003 to rendezvous with the comet 46P/Wirtanen in 2011.^[43] This plan was abandoned after the failure of an Ariane 5 ECA carrier rocket during Hot Bird 7's launch on 11 December 2002, grounding it until the cause of the failure could be determined.^[51] In



Illustration of *Rosetta* and *Philae* at the comet

May 2003, a new plan was formed to target the comet 67P/Churyumov-Gerasimenko, with a revised launch date of 26 February 2004 and comet rendezvous in 2014.^{[52][53]} The larger mass and the resulting increased impact velocity made modification of the landing gear necessary.^[54]

Launch

After two scrubbed launch attempts, *Rosetta* was launched on 2 March 2004 at 07:17 UTC from the Guiana Space Centre in French Guiana, using Ariane 5 G+ carrier rocket.^[3] Aside from the changes made to launch time and target, the mission profile remained almost identical. Both co-discoverers of the comet, *Klim Churyumov* and *Svetlana Gerasimenko*, were present at the spaceport during the launch.^{[55][56]}

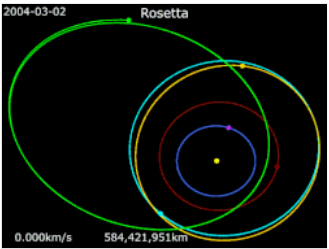
Deep space manoeuvres

To achieve the required velocity to rendezvous with 67P, *Rosetta* used gravity assist manoeuvres to accelerate throughout the inner Solar System.^[16] The comet's orbit was known before *Rosetta*'s launch, from ground-based measurements, to an accuracy of approximately 100 km (62 mi). Information gathered by the onboard cameras beginning at a distance of 24 million kilometres (15,000,000 mi) were processed at ESA's Operation Centre to refine the position of the comet in its orbit to a few kilometres. The first Earth flyby was on 4 March 2005.^[57]

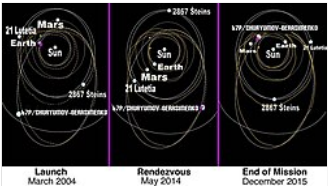
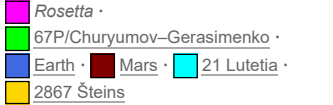
On 4 July 2005, imaging instruments on board observed the collision between the comet Tempel 1 and the impactor of the Deep Impact mission.^[58]

On 25 February 2007, the craft was scheduled for a low-altitude flyby of Mars, to correct the trajectory. This was not without risk, as the estimated altitude of the flyby was a mere 250 kilometres (160 mi).^[59] During that encounter, the solar panels could not be used since the craft was in the planet's shadow, where it would not receive any solar light for 15 minutes, causing a dangerous shortage of power. The craft was therefore put into standby mode, with no possibility to communicate, flying on batteries that were originally not designed for this task.^[60] This Mars manoeuvre was therefore nicknamed "The Billion Euro Gamble".^[61] The flyby was successful, with *Rosetta* even returning detailed images of the surface and atmosphere of the planet, and the mission continued as planned.^{[11][28]}

The second Earth flyby was on 13 November 2007 at a distance of 5,700 km (3,500 mi).^{[62][63]} In observations made on 7 and 8 November, *Rosetta* was briefly mistaken for a near-Earth asteroid about 20 m (66 ft) in diameter by an astronomer of the Catalina Sky Survey and was given the



Animation of *Rosetta*'s trajectory from 2 March 2004 to 9 September 2016



Trajectory of the *Rosetta* space probe

provisional designation 2007 VN₈₄.^[64] Calculations showed that it would pass very close to Earth, which led to speculation that it could impact Earth.^[65] However, astronomer Denis Denisenko recognised that the trajectory matched that of *Rosetta*, which the Minor Planet Center confirmed in an editorial release on 9 November.^{[66][67]}

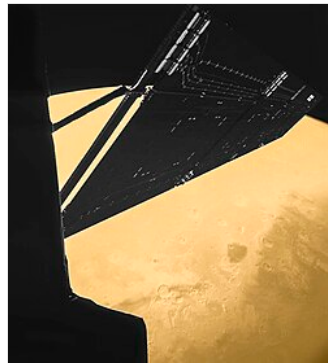
The spacecraft performed a close flyby of asteroid 2867 Šteins on 5 September 2008. Its onboard cameras were used to fine-tune the trajectory, achieving a minimum separation of less than 800 km (500 mi). Onboard instruments measured the asteroid from 4 August to 10 September. Maximum relative speed between the two objects during the flyby was 8.6 km/s (19,000 mph; 31,000 km/h).^[68]

Rosetta's third and final flyby of Earth happened on 12 November 2009 at a distance of 2,481 km (1,542 mi).^[69]

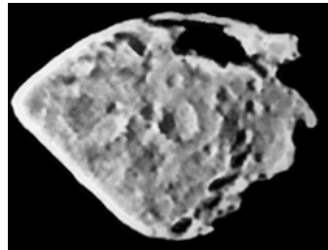
On 16 March 2010, *Rosetta* performed observations of the dust tail of the object P/2010 A2. Together with observations by Hubble Space Telescope, it helped to confirm that P/2010 A2 is not a comet, but an asteroid, and that the tail most likely consists of particles from an impact by a smaller asteroid.^[70]

On 10 July 2010, *Rosetta* flew by 21 Lutetia, a large main-belt asteroid, at a minimum distance of $3,168 \pm 7.5$ km ($1,969 \pm 4.7$ mi) at a velocity of 15 kilometres per second (9.3 mi/s).^[13] The flyby provided images of up to 60 metres (200 ft) per pixel resolution and covered about 50% of the surface, mostly in the northern hemisphere.^{[30][71]} The 462 images were obtained in 21 narrow- and broad-band filters extending from 0.24 to 1 μm .^[30] Lutetia was also observed by the visible–near-infrared imaging spectrometer VIRTIS, and measurements of the magnetic field and plasma environment were taken as well.^{[30][71]}

On 8 June 2011, the spacecraft was transferred into a spin stabilised mode and all electronics except the onboard computer and the hibernation heaters were switched off for the planned 31 months of hibernation.^[72] After leaving its hibernation mode in January 2014 and getting closer to the comet, *Rosetta* began a series of eight burns in May 2014. These reduced the relative velocity between the spacecraft and 67P from 775 to 7.9 m/s (2,543 to 26 ft/s).^[32]



Rosetta "selfie" at Mars



Enhanced image of asteroid Šteins by *Rosetta*



Earth from *Rosetta* during final flyby

Reaction control system problems

In 2006, *Rosetta* suffered a leak in its reaction control system (RCS).^[16] The system, which consists of 24 bipropellant 10-newton thrusters,^[32] was responsible for fine tuning the trajectory of *Rosetta* throughout its journey. The RCS operated at a lower pressure than designed due to the leak. While this may have caused the propellants to mix incompletely and burn 'dirtier' and less efficiently, ESA engineers were confident that the spacecraft would have sufficient fuel reserves to allow for the successful completion of the mission.^[73]

Prior to *Rosetta*'s deep space hibernation period, two of the spacecraft's four reaction wheels began exhibiting increased levels of "bearing friction noise". Increased friction levels in Reaction Wheel Assembly (RWA) B were noted after its September 2008 encounter with asteroid Šteins. Two attempts were made to relubricate the RWA using an on-board oil reservoir, but in each case noise levels were only temporarily lowered, and the RWA was turned off in mid-2010 after the flyby of asteroid Lutetia to avoid possible failure. Shortly after this, RWA C also began showing evidence of elevated friction. Relubrication was also performed on this RWA, and methods were found to temporarily increase its operating temperature to better improve the transfer of oil from its reservoir. In addition, the reaction wheel's speed range was decreased to limit lifetime accumulated rotations. These changes resulted in RWA C's performance stabilising.^[74]

During the spacecraft's Deep Space Hibernation flight phase, engineers performed ground testing on a flight spare RWA at the European Space Operations Centre. After *Rosetta* exited hibernation in January 2014, lessons learned from the ground testing were applied to all four RWAs, such as increasing their operating temperatures and limiting their wheel speeds to below 1000 rpm. After these fixes, the RWAs showed nearly identical performance data.^[74] Three RWAs were kept operational, while one of the malfunctioning RWAs was held in reserve. Additionally, new on-board software was developed to allow *Rosetta* to operate with only two active RWAs if necessary.^{[16][75]} These changes allowed the four RWAs to operate throughout *Rosetta*'s mission at 67P/Churyumov–Gerasimenko despite occasional anomalies in their friction plots and a heavy workload imposed by numerous orbital changes.^[74]

Orbit around 67P

Starting on 7 May 2014, *Rosetta* began orbital correction manoeuvres to bring itself into orbit around 67P. At the time of the first deceleration burn *Rosetta* was approximately 2,000,000 km (1,200,000 mi) away from 67P and had a relative velocity of +775 m/s (2,540 ft/s); by the end of the last burn, which occurred on 23 July, the distance had been reduced to just over 4,000 km (2,500 mi) with a relative velocity of +7.9 m/s (18 mph).^{[32][76]} In total eight burns were used to align the trajectories of *Rosetta* 67P with the majority of the deceleration occurring during three burns: Delta-v's of 291 m/s (650 mph) on 21 May, 271 m/s (610 mph) on 4 June, and 91 m/s (200 mph) on 18 June.^[32]



Joy from receiving first signals from *Rosetta* after waking up on 20 January 2014.

In August 2014, *Rosetta* rendezvoused with the comet 67P/Churyumov–Gerasimenko (67P) and commenced a series of manoeuvres that took it on two successive triangular paths, averaging 100 and 50 kilometres (62 and 31 mi) from the nucleus, whose segments are hyperbolic escape trajectories alternating with thruster burns.^{[33][31]} After closing to within about 30 km (19 mi) from the comet on 10 September, the spacecraft entered actual orbit about it.^{[33][31][34]} This started the Global Mapping Phase.^[6]

The surface layout of 67P was unknown before *Rosetta*'s arrival. The orbiter mapped the comet in anticipation of detaching its lander.^[77] On 14 July 2014, the OSIRIS on-board imaging system returned images of the comet which confirmed its irregular shape.^{[78][79]} By 25 August 2014, five potential landing sites had been determined.^[80] On 15 September 2014, ESA announced Site J, named *Agilkia* in honour of Agilkia Island by an ESA public contest and located on the "head" of the comet,^[81] as the lander's destination.^[82]

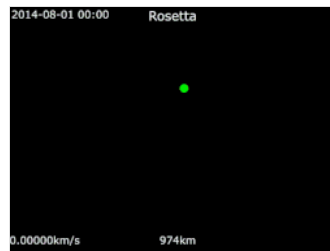
Philae lander

Philae detached from *Rosetta* on 12 November 2014 at 08:35 UTC, and approached 67P at a relative speed of about 1 m/s (3.6 km/h; 2.2 mph).^[83] It initially landed on 67P at 15:33 UTC, but bounced twice, coming to rest at 17:33 UTC.^{[17][84]} Confirmation of contact with 67P reached Earth at 16:03 UTC.^[85]

On contact with the surface, two harpoons were to be fired into the comet to prevent the lander from bouncing off, as the comet's escape velocity is only around 1 m/s (3.6 km/h; 2.2 mph).^[86] Analysis of telemetry indicated that the surface at the initial touchdown site is relatively soft, covered with a layer of granular material about 0.82 feet (0.25 meters) deep,^[87] and that the harpoons had not fired upon landing. After landing on the comet, *Philae* had been scheduled to commence its science mission, which included:

- Characterisation of the nucleus
- Determination of the chemical compounds present, including amino acid enantiomers^[88]
- Study of comet activities and developments over time

After bouncing, *Philae* settled in the shadow of a cliff,^[89] canted at an angle of around 30 degrees. This made it unable to adequately collect solar power, and it lost contact with *Rosetta* when its batteries ran out after three days, well before much of the planned science objectives could be attempted.^{[89][18]} Contact was briefly and intermittently reestablished several months later at various times between 13 June and 9 July, before contact was lost once again. There was no communication



Animation of *Rosetta*'s trajectory around 67P from 1 August 2014 to 31 March 2015

■ *Rosetta* · ■ 67P



Rosetta rendezvous with a comet

afterwards,^[90] and the transmitter to communicate with *Philae* was switched off in July 2016 to reduce power consumption of the probe.^[19] The precise location of the lander was discovered in September 2016 when *Rosetta* came closer to the comet and took high-resolution pictures of its surface.^[89] Knowing its exact location provides information needed to put *Philae*'s two days of science into proper context.^[89]

End of mission

Between November 2014 and December 2015, *Rosetta* escorted the comet around the Sun and performed riskier investigations.^[91] As the orbit of comet 67P took it farther from the Sun, the amount of sunlight reaching *Rosetta*'s solar panels decreased. While it would have been possible to put *Rosetta* into a second hibernation phase during the comet's aphelion, there was no assurance that enough power would be available to run the spacecraft's heaters to keep it from freezing. To guarantee a maximum science return, mission managers made the decision to instead guide *Rosetta* down to the comet's surface and end the mission on impact, gathering photographs and instrument readings along the way.^[92] On 23 June 2015, at the same time as a mission extension was confirmed, ESA announced that end of mission would occur at the end of September 2016 after two years of operations at the comet.^[91]

All stations and the briefing room, we've just had loss of signal at the expected time. This is another outstanding performance by flight dynamics. So we'll be listening for the signal from Rosetta for another 24 hours, but we don't expect any. This is the end of the Rosetta mission. Thank you, and goodbye.

—Sylvain Lodiot, *Rosetta* Spacecraft Operations Manager, European Space Operations Centre^[93]

Rosetta began a 19 km (12 mi) descent with a 208-second thruster burn executed on 29 September 2016 at approximately 20:50 UTC.^{[94][95][93]} Its trajectory targeted a site in the Ma'at region near an area of dust- and gas-producing active pits.^[96]



Comet 67P seen from 10 km (6 mi)



Rosetta and *Philae*



The comet in January 2015 as seen by *Rosetta*'s NAVCAM

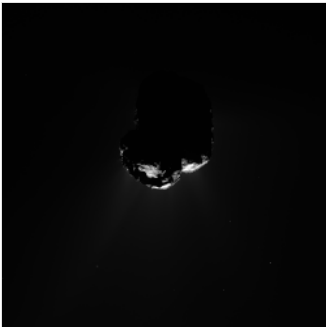
Impact on the comet's surface occurred 14.5 hours after its descent manoeuvre; the final data packet from *Rosetta* was transmitted at 10:39:28.895 UTC (SCET) by the OSIRIS instrument and was received at the European Space Operations Centre in Darmstadt, Germany, at 11:19:36.541 UTC.^{[94][95][97]} The spacecraft's estimated speed at the time of impact was 3.2 km/h (2.0 mph; 89 cm/s),^[21] and its touchdown location, named *Sais* by the operations team after the Rosetta Stone's original temple home, is believed to be only 40 m (130 ft) off-target.^[96] The final complete image transmitted by the spacecraft of the comet was taken by its OSIRIS instrument at an altitude of 23.3–26.2 m (76–86 ft) about 10 seconds before impact, showing an area 0.96 m (3.1 ft) across.^{[96][98]} *Rosetta*'s computer included commands to send it into safe mode upon detecting that it had hit the comet's surface, turning off its radio transmitter and rendering it inert in accordance with International Telecommunication Union rules.^[93]

On 28 September 2017, a previously unrecovered image taken by the spacecraft was reported. This image was recovered from three data packets discovered on a server after completion of the mission. While blurry due to data loss, it shows an area of the comet's surface approximately one square meter in size taken from an altitude of 17.9–21.0 m (58.7–68.9 ft), and represents *Rosetta*'s closest image of the surface.^{[98][99]}

Spacecraft

The *Rosetta* bus was a 2.8 × 2.1 × 2.0 m (9.2 × 6.9 × 6.6 ft) central frame and aluminium honeycomb platform. Its total mass was approximately 3,000 kg (6,600 lb), which included the 100 kg (220 lb) *Philae* lander and 165 kg (364 lb) of science instruments.^[100] The Payload Support Module was mounted on top of the spacecraft and housed the scientific instruments, while the Bus Support Module was on the bottom and contained spacecraft support subsystems. Heaters placed around the spacecraft kept its systems warm while it was distant from the Sun. *Rosetta*'s communications suite included a 2.2 m (7.2 ft) steerable high-gain parabolic dish antenna, a 0.8 m (2.6 ft) fixed-position medium-gain antenna, and two omnidirectional low-gain antennas. The high-gain antenna also had the ESA logo printed on the reflector as a form of branding to let people clearly identify it as project from the European Space Agency (ESA) ^[101]

Electrical power for the spacecraft came from two solar arrays totalling 64 square metres (690 sq ft).^[102] Each solar array was subdivided into five solar panels, with each panel being 2.25 × 2.736 m (7.38 × 8.98 ft). The individual solar cells were made of silicon, 200 μm thick, and 61.95 × 37.75 mm (2.44 × 1.49 in).^[103] The solar arrays generated a maximum of approximately 1,500



Outbursting of the comet 67P/Churyumov–Gerasimenko on 12 September 2015 one of the most dramatic cliff collapses captured during the Rosetta mission.



Comet 67P with a tail of gas and dust, seen from 162 km (101 mi)